

CHAPTER 13

Momentum control

13.1. Space story

The first-shift attitude control engineers had left the momentum-unloading system running at shift change. This system used thrusters to remove pitch momentum from the momentum wheel. The momentum-unloading software was not smart enough to stop when the wheel was unloaded. The operator was on a break and it continued to fire thrusters that knocked the bias-momentum axis off orbit normal and induced large nutation. I got the operator back on the console and we began a series of thruster firings to fix the offset and damp nutation. This was done by phasing the thruster firings with the gyro measurements. As we were getting it under control, the customer called and asked if everything was alright. We responded, “Everything is just fine.”

13.2. Introduction

Momentum unloading is used on spacecraft that have momentum-exchange devices for attitude control. Momentum-exchange devices include reaction wheels and a control-moment gyro. Momentum-exchange devices only exchange momentum between the device and the spacecraft, they can never remove momentum from the spacecraft. If there is a secular, that is an inertial-frame fixed torque, the momentum-exchange devices will eventually run out of momentum storage and need to be unloaded. To unload the momentum, you need an external torque. This can be done with magnetic torquers if your planet has a magnetic field, thrusters, solar pressure, or aerodynamic forces.

In this chapter, we break the problem into momentum growth, the control algorithm, and then torque generation.

13.3. Momentum growth

It is easiest to look at the inertial frame dynamical equations that are simply

$$T = \dot{H} \quad (13.1)$$

where T is the disturbance torque in the inertial frame and H is the inertial spacecraft momentum. Assume that the torque is a combination of a steady torque and a sinusoidal torque

$$T = T_0 + T_s \sin \omega_o t \quad (13.2)$$

where t is time and ω_o is the orbital rate of the satellite. This is typical for a planet-pointing satellite. If the initial momentum is zero we get

$$H = T_0 t + \frac{T_s}{\omega_o} (1 - \cos \omega_o t) \quad (13.3)$$

The momentum will grow linearly with time and oscillate at the orbit rate. If the momentum-exchange devices have sufficient momentum-storage capability, the orbit-rate momentum will be absorbed and released over the orbit. The linear momentum growth will need to be unloaded.

13.4. Control algorithms

One approach is to use a Proportional-Integral (PI) controller of the form

$$T = -k \left(H + \frac{1}{\tau_I} \int H \right) \quad (13.4)$$

The first term generates a torquer proportional to the momentum. The second produces a torque proportional to the integral of the momentum. This drives the error to zero, without any offset. We write the equations of motion with the control ω for the control is set to be ten times orbit rate.

$$T_d = \dot{H} + k \left(H + \frac{1}{\tau_I} \int H \right) \quad (13.5)$$

where T_d is the disturbance torque. Integrate once

$$H_d = \ddot{H} + k\dot{H} + \frac{k}{\tau_I} H \quad (13.6)$$

which is a second-order damped oscillator. The controller gains are related to the damping ratio and the undamped natural frequency by

$$k = 2\zeta\omega \quad (13.7)$$

$$\tau_I = \frac{2\zeta}{\omega} \quad (13.8)$$

Typically, you want your controller to be faster than the disturbance frequency. The damping ratio, ζ should be greater than 0.7071, the critical damping ratio. One is often chosen for ζ since overshoot is undesirable.

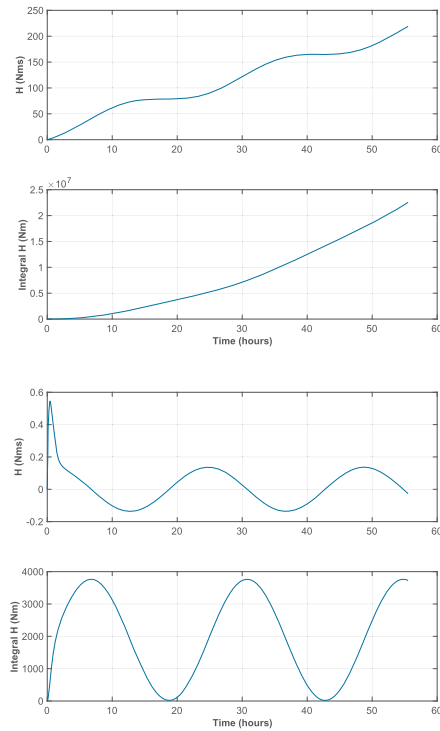
Example 13.1 shows the results with and without control for a single axis. The control system is built into the right-hand-side dynamical model and can be turned on and off via the data structure. When the control is off, the momentum grows linearly,

as expected with an oscillatory component. When the control is on, the integral term drives the offset to zero.

```

1 cOn = true; % Control is on if true
2 x = [0;0]; % [H;integral H]
3 n = 2000; % Number of steps
4 dT = 100; % Seconds
5 omega = 2*pi/8640;
6 zeta = 1;
7
8 xP = zeros(2,n); % Store plot data
9 d = struct('k',2*zeta*omega,...
10           'tau',2*zeta/omega,...
11           'tD',0.001,'tS',0.0001,...
12           'omega',2*pi/86400);
13 if (~cOn) % Set the forward gain
14     d.k = 0;
15 end
16 t = (0:n-1)*dT;
17 for k = 1:n
18     xP(:,k) = x;
19     x = RK4(@RHSMomentum,x,dT,t(k),d);
20 end
21 yL = {'H_(Nms)', 'Integral_H_(Nm)'};
22 TimeHistory(t,xP,yL,'Momentum');
23
24 %% Right hand side
25 function xDot = RHSMomentum(x,t,d)
26 f = d.tD + d.tD*sin(d.omega*t) ...
27     - d.k*(x(1) + x(2)/d.tau);
28 xDot = [f;x(1)];
29
30 end

```



Example 13.1: Momentum control. The right-hand side for the momentum dynamical equations is a function at the bottom of the listing.

13.5. Control-torque generation

13.5.1 Thruster control

13.5.1.1 Direct

If we are using thrusters for momentum unloading we just transform the inertial torque demand into the body frame

$$T_B = AT_{ECI} \quad (13.9)$$

where A is a transformation matrix from the inertial to the body frame. This can be found in your attitude quaternion. It assumes you have three-axis information. The torque generated by the thrusters is

$$T_B = \sum_i (r_i - c) \times F_i \quad (13.10)$$

where F is the thruster force vector, r_i is the location of the i th thruster with respect to the reference point and c is the location of the center-of-mass with respect to the same reference frame. The control law given in the last section is linear and proportional. This means the torque demand, and therefore the force demand will go from zero to the maximum force the thruster can provide. You may note that one thruster cannot provide any three-axis torque because $(r_i - c)^\times$ for just one thruster is not invertible. A sum of thrusters will be as long as they have moment arms so that three-axis torques are possible.

Small thrusters are rarely throttleable. Therefore, we approximate a linear thruster by using pulsewidth modulation. Pulsewidth modulation is turning on a thruster for only a portion of the control period. If the thruster has a force F_{max} and we want the average force over the control period, τ to be F we just turn the thruster on for part of the control period

$$\tau_i = \frac{F_i}{F_{max}} \tau \quad (13.11)$$

where τ_i will have a minimum. You do not need to make τ the same length as your attitude control period. Since momentum unloading is usually slow you can make it much lower than your control period, which gives your momentum-unloading system a much higher resolution.

13.5.1.2 Off-pulsing

Off-pulsing is a method of getting attitude control from thrusters that are normally on. Suppose you have thrusters on the four corners of your spacecraft that are turned on to get thrust in that body's direction. For example, in a geosynchronous satellite, you might use four East-faced thrusters for an East-West station-keeping maneuver. If the thrusters are oriented and located so that you do not get a net torque, the attitude is undisturbed. The situation is shown in Fig. 13.1. Off-pulsing is exactly like onpulsing. Instead of turning on a thruster for the pulsewidth, you delay turning on a thruster for the computed pulsewidth duration. This puts the torque generation closest to the sensor measurement. If you instead, shut the pulse off early, you would have almost a pulse-period delay. The sign of the torque is flipped for the thruster selection. The thruster that has the longest pulsewidth produces less torque than the opposite thruster.

13.5.2 Magnetic control

13.5.2.1 Magnetic field

If the planet about which you are orbiting has a magnetic field, then you can use magnetic torquers for momentum control. Planets and moons with significant magnetic fields are

- Earth;

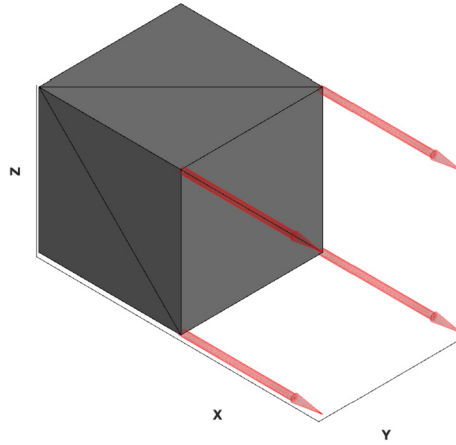


Figure 13.1 Four thrusters on for a velocity-change maneuver. The arrows show the plume direction.

- Jupiter;
- Saturn;
- Uranus;
- Neptune;
- Ganymede;
- Europa.

The simplest approximation to a planet's magnetic field is a magnetic dipole oriented along the magnetic axis, which is not necessarily the rotation axis. Example 13.2 shows a dipole model of the Earth in three different orbits, equatorial, ISS, and polar. The field is given in the ECI frame. In the polar and equatorial orbits, one component of the magnetic field is constant. This model is a dipole. The magnetic field is also affected by the Sun and that may have to be considered, especially for high orbits.

13.5.2.2 Instantaneous control

We transform the inertia frame torque demand into the body frame

$$T_B = A^T T_{\text{ECI}} \quad (13.12)$$

where A is the transformation matrix from the body to the ECI frame. A is known from the measured attitude quaternion. The torque is produced by magnetic torquers

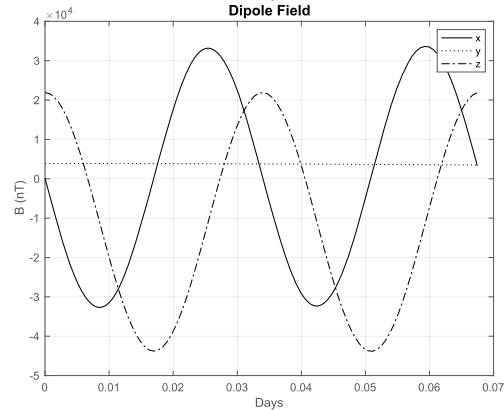
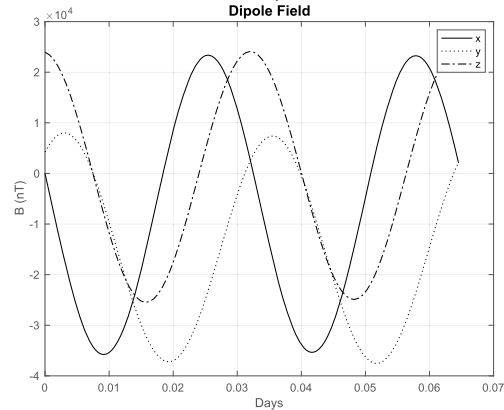
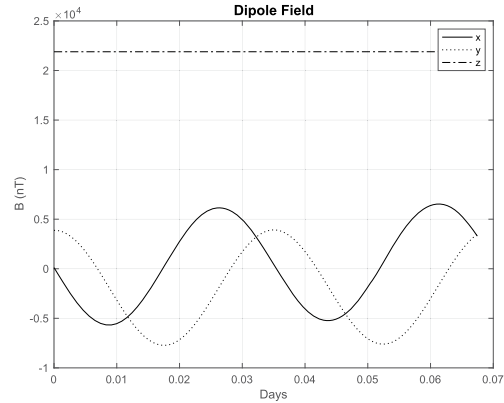
$$T_B = M \times B \quad (13.13)$$

where M is the magnetic torquer dipole and B is the magnetic field measured by a magnetometer or from an onboard magnetic-field model if we know the orbital position

```

1 %% ISS
2 [el,jD0] = ISSOrbit;
3 % Returns t for an orbit period
4 [r,~,t] = RVOrbGen(el);
5 % Julian date is for the rotation of the Earth
6 jD = jD0 + t/86400;
7 BDipole(r,jD)
8
9 %% Polar
10 % Elements are [semi-major axis, inclination ...]
11 [r,~,t] = RVOrbGen([7000 pi/2 0 0 0 0]);
12 jD = jD0 + t/86400;
13 BDipole(r,jD)
14
15 %% Equatorial
16 [r,~,t] = RVOrbGen([7000 0 0 0 0 0]);
17 jD = jD0 + t/86400;
18 BDipole(r,jD)

```



Example 13.2: The Earth's magnetic field in equatorial, ISS, and polar orbits. The equatorial and polar orbits are at radii of 7000 km.

accurately. For some orbits, like geosynchronous, the magnetic field is fairly constant so measurements are not required. B is in the body frame. Flip the order of M and B using a vector identity

$$T_B = -B^\times M \quad (13.14)$$

Since the first term is skew-symmetric, it cannot be inverted to find M . Instead, we do a least-squares fit.

$$M = \frac{B^\times T_B}{|B|^2} \quad (13.15)$$

The simplest way to implement the unloading is to pulsewidth modulate the magnetic torquers, just like with thrusters. A magnetic torquer can then be always maximum on or off. It is controlled by a switch. If you wanted a variable dipole, you would need to add a linear amplifier, which would waste energy. The magnetic-torquer pulse period is selected that is n times as long as the control period because unloading can be a slower process than control. Let $M_{\max}$ be the saturated value of the dipole. The fraction of the time on is

$$f = \frac{|M|}{M_{\max}} \quad (13.16)$$

The torquer coil is on for n_{on} cycles

$$n_{\text{on}} = \text{round}(fn) \quad (13.17)$$

The polarity is the sign of M .

Example 13.3 shows the transient response of a magnetic-torquer system. It is removing momentum stored at the beginning of the simulation. Only proportional control is used. The simulation compares the magnetic-torquer control with an ideal proportional controller. The spacecraft is in an ISS orbit. The dipole moment is determined by the forward gain, k .

The ideal torque reduces the momentum smoothly. The magnetic control eventually removes all of the momentum but increases it on the x -axis at the beginning.

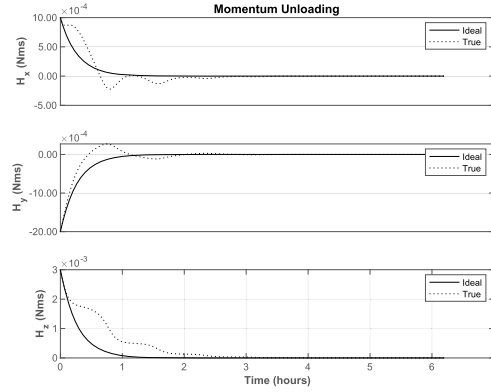
13.5.2.3 Individual torquer control

Three torquers are not needed to unload momentum. Another approach is to find the component of the torquer demand along a particular magnetic-torquer torque direction, set the dipole to the fraction of the dipole moment that produces that torque, subtract

```

1  secInDay = 86400;
2  n = 1000;
3  [el,jD0] = ISSOrbit;
4  jDStart = Date2JD;
5  nOrbits = 4;
6  gain = 0.001; % Unloading gain
7
8  %% Orbit
9  p = Period(el(1));
10 t = linspace(0,nOrbits*p,n);
11 jD = jD0 + t/secInDay;
12 [r,v] = RVOrbGen(el,t);
13
14 %% Magnetic field
15 b = BDipole(r,jD);
16
17 % Set up the plotting arrays
18 h0 = [1;-2;3]*1e-3;
19 hECI = zeros(3,n);
20
21 % Orients the spacecraft with local
22 % vertical and local horizontal
23 q = QVLH(r,v);
24 b = QForm(q,b);
25 h = QForm(q(:,1),h0);
26 hECI(:,1) = h0;
27 hECIM = hECI;
28 mDipole = zeros(3,n);
29 dT = t(2); % Time step
30
31 %% Simulate unloading a constant momentum
32 for k = 2:n
33     sB = Skew(b(:,k-1));
34     tMM = gain*hECIM(:,k-1);
35     mDipole(:,k-1) = pinv(sB*sB)*sB*tMM;
36     tIdeal = -gain*hECI(:,k-1);
37     hECI(:,k) = hECI(:,k-1) + dT*tIdeal;
38     mB = Cross(mDipole(:,k-1),b(:,
39         k-1));
40     hECIM(:,k) = hECIM(:,k-1) + dT*mB;
41 end
42 TimeHistory(t,[hECI;hECIM],...
43     {'Hx_u(Nms)' 'Hy_u(Nms)' 'Hz_u(Nms)'} ,...
44     'Momentum_Unloading', {[1 4] [2 5] [3 6]} ,...
45     [{'Ideal' 'True'} {'Ideal' 'True'} {'Ideal' '
True'}]);

```



Example 13.3: Proportional-control momentum unloading. Shows ideal proportional control and control with magnetic torquers.

from the torque demand, and then cycle through the rest of the torquers.

$$T_M = M^\times B \quad (13.18)$$

$$u_M = \frac{T_M}{|T_M|} \quad (13.19)$$

$$T_C = (T_B^T u_M) u_M \quad (13.20)$$

$$m = \frac{|T_C|}{|T_M|} \quad (13.21)$$

$$T_B = T_B - T)C \quad (13.22)$$

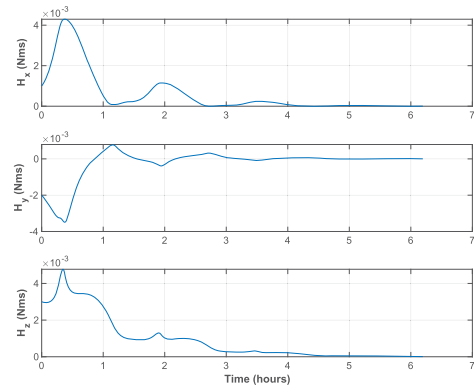
where m is the dipole fraction. You repeat this loop for each torquer.

Example 13.4 shows the transient response of a magnetic-torquer system with only two torquers. It uses the same control algorithm as the previous example.

```

1 % The torquers
2 m      = [1 0;0 1;0 0];
3 nM      = size(m,2);
4
5 secInDay = 86400;
6 n        = 1000;
7 [el,jD0] = ISSOrbit;
8 jDStart   = Date2JD;
9 nOrbits   = 4;
10 gain     = 0.001; % Unloading gain
11
12 %% Orbit
13 p        = Period(el(1));
14 t         = linspace(0,nOrbits*p,n);
15 jD        = jD0 + t/secInDay;
16 [r,v]     = RVOrbGen(el,t);
17
18 %% Magnetic field
19 b         = BDipole(r,jD);
20
21 % Initial momentum
22 h0        = [1;-2;3]*1e-3;
23
24 % Set up the plotting array
25 hECI      = zeros(3,n);
26
27 % Orients the spacecraft with local
28 % vertical and local horizontal
29 q         = QVLH(r,v);
30 b         = QForm(q,b);
31 h         = QForm(q(:,1),h0);
32 hECI(:,1) = h0;
33 mDipole   = zeros(3,n);
34 dT        = t(2); % Time step
35
36 %% Simulate unloading a constant momentum
37 for k = 2:n
38     tD = gain*hECI(:,k-1);
39     mD = zeros(3,1);
40
41     % Loop through the torquers
42     for j = 1:nM
43         tM = Cross(m(:,j),b(:,k-1));
44         uTM = Unit(tM);
45         tCK = (tD'*uTM)*uTM;
46         mD = mD + Mag(tCK)*m(:,j)/Mag(tM);
47         tD = tD - tCK;
48     end
49     mB = Cross(mD,b(:,k-1));
50     hECI(:,k) = hECI(:,k-1) + dT*mB;
51 end
52
53 TimeHistory(t,hECI,{ 'H_x_{\perp}(Nms)' 'H_y_{\perp}(Nms)' 'H_z_{\perp}(Nms)' },...
54 'Momentum_Unloading');

```



Example 13.4: Individual torquer-momentum unloading. Shows ideal proportional control and control with two magnetic torquers.

Ideally, the momentum-control torque reduces the momentum smoothly. In this example, the magnetic control eventually removes all of the momentum but does increase the momentum on the x -axis at the beginning of the momentum-unloading operation.

13.5.2.4 Average control

While a magnetic torquer cannot produce a specific three-axis torque at a given time, it may be able to produce an average three-axis torque that meets the demand if the magnetic field varies sufficiently over time.

The torque produced by a torquer of dipole m is

$$T = M^\times B \quad (13.23)$$

where the skew-symmetric matrix for M is

$$M^\times = \begin{bmatrix} 0 & -M_z & M_y \\ M_z & 0 & -M_x \\ -M_y & M_x & 0 \end{bmatrix} \quad (13.24)$$

The above matrix cannot be inverted; consequently, it is not possible to realize a three-axis torque demand with magnetic torquers. However, if B changes it will be possible on average to satisfy an average three-axis demand. Assume that we want to drive the momentum vector h to zero over an interval T . Write the Hamiltonian, which is the cost you want to minimize. This applies a cost to the quadratic of the dipole and adjoins the right-hand-side of the dynamical equations

$$\mathcal{H} = \frac{1}{2} M^T M - \lambda^T B^\times M \quad (13.25)$$

where λ are the Lagrange multipliers that must be found as part of the solution. The control law is found by taking the derivative of $\mathcal{H}$ with respect to m and setting the result to zero

$$M = -B^\times \lambda \quad (13.26)$$

where λ is a constant vector. Therefore, substituting the above into the constraint equation gives

$$h = \int_0^T B^\times M dt \quad (13.27)$$

$$h = - \left[\int_0^T B^\times B^\times dt \right] \lambda \quad (13.28)$$

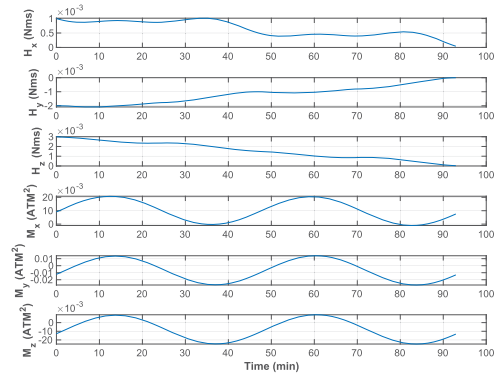
$$\lambda = \left[\int_0^T B^\times B^\times dt \right]^{-1} h \quad (13.29)$$

If the integral is invertible, it is possible to drive h to zero over the interval T . It is necessary to know B over the time interval T , which requires an onboard ephemeris or recording magnetometer readings from the previous orbit. Example 13.5 shows average

control. The momentum is driven to zero over one orbit. The starting momentum is the same as in Example 13.3. That controller reduces the momentum to near zero in one orbit but takes several to completely zero out the momentum.

```

1 h0      = [1;-2;3]*1e-3;
2
3 %% ISS
4 [el,jD0] = ISSOrbit;
5
6 % Returns t for an orbit period
7 [r,~,t] = RVOrbGen(el);
8 n       = length(t);
9
10 % Julian date is for the rotation of the Earth
11 jD      = jD0 + t/86400;
12 b       = BDipole(r,jD);
13 f       = zeros(3,3);
14 dT      = t(2);
15 for k = 1:n
16     s = Skew(b(:,k));
17     f = f + s*s;
18 end
19 lambda = dT*f\h0;
20 h       = zeros(3,n);
21 h(:,1) = h0;
22 m       = Cross(lambda,b);
23 for k = 1:n-1
24     h(:,k+1) = h(:,k) - dT*Cross(m(:,k),b(:,k));
25 end
26 yL = {'H_x\(\text{Nms})' 'H_y\(\text{Nms})' 'H_z\(\text{Nms})' ,...
27       'M_x\(\text{ATM}^2)' 'M_y\(\text{ATM}^2)' 'M_z\(\text{ATM}^2)'
28       };
29 TimeHistory(t,[h;m],yL,'Momentum')
```



Example 13.5: Average control.

13.5.3 Solar and aerodynamic pressure

13.5.3.1 Introduction

Solar and aerodynamic forces put torques on the spacecraft. They are major sources of disturbances and momentum growth. They can also be used as actuators. Normally, one would not add surfaces for control but it is possible to take advantage of either type of force for momentum management. Two examples will be given. One is establishing a torque-equilibrium attitude using aerodynamic forces. The other is using the solar wings on a satellite for momentum control.

13.5.3.2 Torque-equilibrium attitude

If your spacecraft has the flexibility to change its orientation, it may be possible to fly it in a torque-equilibrium attitude to balance out the disturbance torques. The most common situation is to balance gravity-gradient and aerodynamic torques in low-Earth orbit about the Earth-pointing axis, which is the LVLH $+z$ or yaw axis. The gravity-gradient torque in the body frame is

$$t_g = \mu \frac{r_b^\times I r_b}{|r_b|^5} \quad (13.30)$$

where I is the inertia matrix, μ is the gravitational constant, and r_b is the position vector of the spacecraft in the body frame. r_b is a function of the spacecraft attitude. The aerodynamic torque is the sum of the aerodynamic forces on all of the elements of the spacecraft that are not shadowed for the gas flow around the spacecraft. The force on each element is opposite the velocity vector. As we showed in the disturbances chapter, the actual force vector may have a component transverse to the normal, but that would not change the analysis. Defined the dynamic pressure as

$$p = \frac{1}{2} \rho |v|^2 \quad (13.31)$$

where ρ is the atmospheric density and v is the velocity. The aerodynamic torque is

$$t_a = -p \sum_i C_{D_i} (n_i^T u_b) a_i (r_i - c) \times u_b \quad (13.32)$$

where C_{D_i} is the drag coefficient of plate i , a_i is the area of plate i , n_i is the surface normal of the plate in the body frame, c is the location of the center-of-mass, and u_b is

$$u_b = \frac{v_b}{|v_b|} \quad (13.33)$$

The geometry is shown in Fig. 13.2. We want to find an attitude where

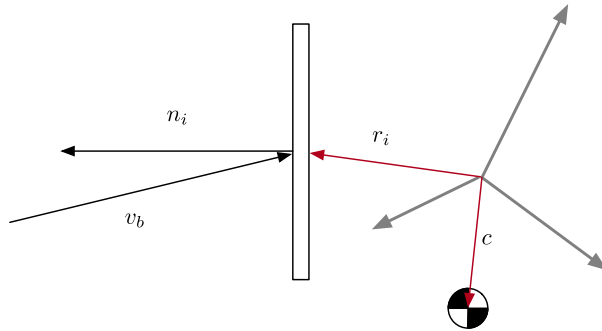


Figure 13.2 Aerodynamics geometry.

$$t_a + t_g = 0 \quad (13.34)$$

We can explore this with a simple case. Assume our spacecraft is a single plate with its normal along the velocity vector, which is the y -axis. The position vector is along the body z -axis. The center-of-mass is at zero and the plate is offset along the x -axis. The geometry is shown in Fig. 13.3. When the spacecraft is aligned with the LVLH frame there is no aerodynamic force on the spacecraft. Eq. (13.34) becomes

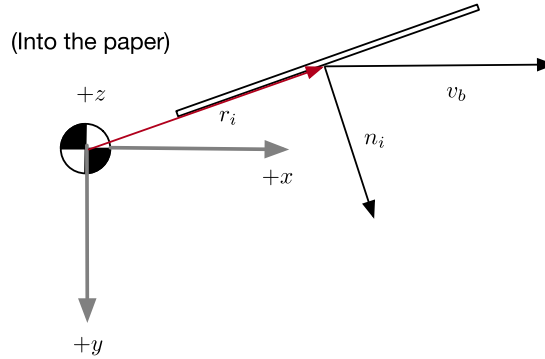


Figure 13.3 Simplified aerodynamics geometry.

$$t_a + \omega_o^2 u^\times I u = 0 \quad (13.35)$$

where u is the unit orbit vector in the body frame and x is the x coordinate of the plate moment arm. The gravity-gradient term was simplified using the orbit rate

$$\omega_o^2 = \frac{\mu}{|r|^2} \quad (13.36)$$

When the position vector is aligned with the body z -axis the gravity-gradient torque is

$$t_g = \omega_o^2 \begin{bmatrix} -I_{yz} \\ I_{xz} \\ 0 \end{bmatrix} \quad (13.37)$$

We can compensate for this disturbance by tilting the spacecraft. If the angles are small, the transformation matrix from LVLH to the body is

$$A = \begin{bmatrix} 1 & \gamma & -\beta \\ -\gamma & 1 & \alpha \\ \beta & -\alpha & 1 \end{bmatrix} \quad (13.38)$$

The angles and signs can be found by inspection by looking at Fig. 13.4. The unit position and velocity vectors in the body frame are

$$u = \begin{bmatrix} -\beta \\ \alpha \\ 1 \end{bmatrix} \quad u_b = \begin{bmatrix} 1 \\ -\gamma \\ \beta \end{bmatrix} \quad (13.39)$$

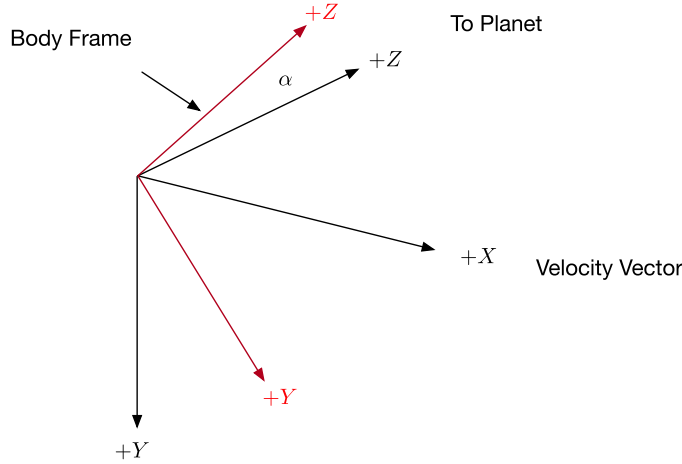


Figure 13.4 Coordinate transformations from LVLH to body with small angles. This diagram shows the roll angle, α . The body axes are in red.

The unit vectors and the transformation matrix are approximations. The vectors are not exactly unit vectors and the transformation matrix is not exactly orthonormal. If the angles are small, they are sufficiently accurate for a linear approximation.

The linearized aerodynamic torque is

$$t_a = -pC_D a \left(\begin{bmatrix} 0 \\ n_x r_z \\ -n_x r_y \end{bmatrix} + \begin{bmatrix} 0 & r_y n_x & r_z n_x \\ 0 & r_z n_z - r_x n_x & -r_z n_y \\ 0 & -r_y n_z & r_y n_y - r_x n_x \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \right) \quad (13.40)$$

The linearized gravity-gradient torque is

$$t_g = -\omega_o^2 \left(\begin{bmatrix} -I_{yz} \\ I_{xz} \\ 0 \end{bmatrix} + \begin{bmatrix} I_{zz} - I_{yy} & I_{xy} & 0 \\ I_{xy} & I_{zz} - I_{xx} & 0 \\ -I_{xz} & -I_{yz} & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \right) \quad (13.41)$$

The angle vector can be solved from these equations. Let b be the sum of the matrix parts and c the sum of the vector parts

$$b = \omega_o^2 \begin{bmatrix} I_{zz} - I_{yy} & I_{xy} & 0 \\ I_{xy} & I_{zz} - I_{xx} & 0 \\ -I_{xz} & -I_{yz} & 0 \end{bmatrix} - pC_D a \begin{bmatrix} 0 & r_y n_x & r_z n_x \\ 0 & r_z n_z - r_x n_x & -r_z n_y \\ 0 & -r_y n_z & r_y n_y - r_x n_x \end{bmatrix} \quad (13.42)$$

and

$$c = \begin{bmatrix} -I_{yz} \\ I_{xz} \\ 0 \end{bmatrix} - p C_D a \begin{bmatrix} 0 \\ n_x r_z \\ -n_x r_y \end{bmatrix} \quad (13.43)$$

The angles are found by solving the following equation.

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = b^{-1} c \quad (13.44)$$

where b must be invertible. Example 13.6 solves the equations for the simple geometry. An alternative approach would be to use a search algorithm. This would retain all the

```

1 i = [2.5 0.2 0.1;0.2 2.2 0.1;0.1 0.1 0.5]*1e
   -3;
2 n = [0;1;0];
3 mu = 3.98600436e5;
4 sMA = 7000;
5 v = sqrt(mu/sMA);
6 omega = sqrt(mu/sMA^3);
7 omSq = omega^2;
8 a = 1;
9 cD = 2.7;
10 r = [0.1;0.2;-0.1];
11 rho = AtmDens2(sMA-6378.165);
12 p = 0.5*rho*cD*v^2;
13
14 tAC = -p*a*[0;n(1)*r(3);-n(1)*r(2)];
15 aA = -p*a*[0 n(1)*r(2) n(1)*r(3)
   ;... 0 n(3)*r(3)-n(1)*r(1) -n(2)*r(3)
   ;... 0 -r(2)*n(3) r(2)*n(2)-r
   (1)*n(1)];
16
17
18
19 aG = omSq*[i(3,3)-i(2,2) i(1,2) 0;i(1,3) i
   (3,3)-i(1,1) 0;-i(1,3) -i(2,3) 0];
20
21 tGC = omSq*[-i(2,3) i(1,3);0];
22
23
24 ang = -(aG + aA)\(tAC + tGC);

```

```

1 ang =
2
3 -0.0533
4 0.0472
5 0.4289

```

Example 13.6: Torque-equilibrium attitude solution.

nonlinear terms. However, it would not provide much intuition. The result of the linear equation could be used as a starting point for a search.

13.5.3.3 Momentum management with solar wings

If a spacecraft has two solar wings, which is true for most geosynchronous spacecraft, they can be used to remove roll and yaw momentum. Roll is the axis pointing at the Earth and yaw is along the velocity vector. The geometry is shown, for one wing, in Fig. 13.5.

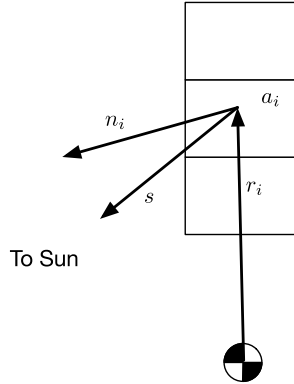


Figure 13.5 Solar-pressure unloading geometry.

If we assume that the solar wings rotate about pitch, γ , and are perfectly flat, then the torque due to the two wings is

$$t = -p \sum_{i=2} N a_i r_i^{\times} \left(s^T n_i (2(\rho_s + \rho_d/3) n_i + (\rho_a + \rho_d) s) \right) \quad (13.45)$$

where p is the solar pressure, which is the solar flux divided by the speed of light, a_i is the panel area, assumed to be the same for both wings, s is the Sun vector, n is the unit normal to the surface, A is the area of the surface, S is the solar flux in N/m^2 , ρ_s is the specular reflection coefficient, ρ_a is the absorbed radiation coefficient, and ρ_d is the diffuse reflection coefficient. The center-of-mass is assumed to be at the origin.

Example 13.7 shows the effect of rotating the solar wings together or separately. The torques and angles are at the top of each figure.

Some spacecraft add a tab at the base of the wing that is perpendicular to the wing plane. That can easily be accommodated in this analysis by adding additional panels with the appropriate areas, normals, and position vectors.

13.5.3.4 Gravity-gradient momentum management

A gravity gradient can also be used to unload momentum. The gravity-gradient torque is

$$T_x = \omega_o^2 [(I_{zz} - I_{yy})\alpha + I_{xy}\beta - I_{yz}] \quad (13.46)$$

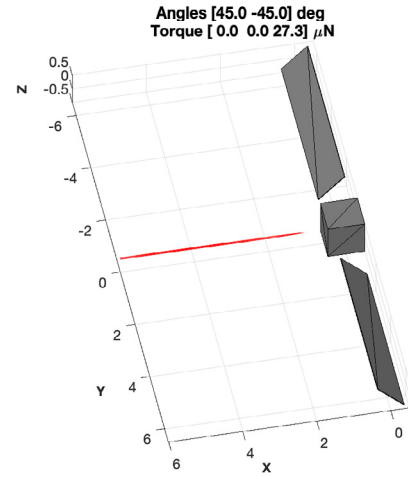
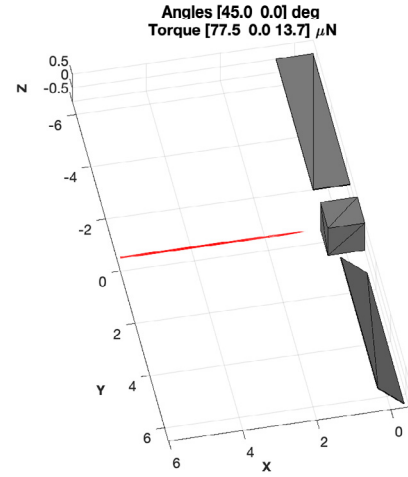
$$T_y = \omega_o^2 [(I_{zz} - I_{xx})\alpha + I_{xy}\beta + I_{xz}] \quad (13.47)$$

$$T_z = -\omega_o^2 [I_{xz}\alpha + I_{yz}\beta] \quad (13.48)$$

```

1 p      = 1367/3e8;
2 rho    = [0.3; 0.7; 0];
3 u      = [0;0;1];
4
5 ang     = [45 45;...
6           0 -45];
7
8 l       = 5; % Wing length (m)
9 w       = 1; % Wing width (m)
10
11 % Vertices and faces for drawing
12 [vP,fP] = Box(w,l,0.02);
13 [vB,fB] = Box(1,1,1);
14
15 % Position vectors
16 rP      = [0; 1/2+1.5;0];
17 rN      = [0;-1/2-1.5;0];
18
19 area    = w*l;
20
21 torque  = zeros(3,size(ang,2));
22
23 for k = 1:size(ang,2)
24     bP    = [CosD(ang(1,k)) 0 -SinD(ang(1,k));...
25             0 1 0;...
26             SinD(ang(1,k)) 0 CosD(ang(1,k))];
27
28     bN    = [CosD(ang(2,k)) 0 -SinD(ang(2,k));...
29             0 1 0;...
30             SinD(ang(2,k)) 0 CosD(ang(2,k))];
31
32     forceP = SolarF(p,rho,u,bP'*u,area);
33     forceN = SolarF(p,rho,u,bN'*u,area);
34     torque(:,k) = Cross(rP,forceP) +...
35                 Cross(rN,forceN);
36
37 % For drawing
38 v      = vB;
39 f      = fB;
40
41 f      = [f;fP+size(v,1)];
42 vW(:,2) = vW(:,2) + 4;
43 vR      = (bP*vW')';
44 v      = [v;vR];
45 f      = [f;fP+size(v,1)];
46 vW(:,2) = vW(:,2) - 4;
47 vR      = (bN*vW')';
48 v      = [v;vR];
49
50
51 s1 = sprintf('Angles [%4.1f %4.1f] deg\n',ang
52             (:,k));
53 s2 = sprintf('Torque [%4.1f %4.1f %4.1f] \muN',
54             torque(:,k)*1e6);
55
56 DrawObject(v,f,[s1 s2]);
57 view([0.2 1 1]);
58 title(s);
59 end

```



Example 13.7: Solar-pressure momentum control.

In matrix form

$$\omega_0^2 \begin{bmatrix} I_{zz} - I_{yy} & I_{xy} & 0 \\ I_{zz} - I_{xx} & I_{xy} & 0 \\ -I_{xz} & -I_{yz} & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} T_x + \omega_0^2 I_{yz} \\ T_y - \omega_0^2 I_{xz} \\ T_z \end{bmatrix} \quad (13.49)$$

It is not possible to unload T_z . We can unload x and y ,

$$\omega_0^2 \begin{bmatrix} I_{zz} - I_{yy} & I_{xy} \\ I_{zz} - I_{xx} & I_{xy} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} T_x + \omega_0^2 I_{yz} \\ T_y - \omega_0^2 I_{xz} \end{bmatrix} \quad (13.50)$$

Analytically, invert the equation and we get

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{\begin{bmatrix} I_{xy} & -I_{xy} \\ I_{xx} - I_{zz} & I_{zz} - I_{yy} \end{bmatrix}}{\omega_0^2 I_{xy}(I_{xx} - I_{zz})} \begin{bmatrix} T_x + \omega_0^2 I_{yz} \\ T_y - \omega_0^2 I_{xz} \end{bmatrix} \quad (13.51)$$

where x and y momentum can be unloaded as long as the denominator is not zero. Example 13.8 gives an example.

```
1 % 3U inertia
2 i = [2.5 0.2 0.1; 0.2 2.5 0.1; 0.1 0.1 0.5]*1e-3
3 mu = 3.98600436e5;
4 sMA = 7000;
5 omega = sqrt(mu/sMA^3)
6 omSq = omega^2;
7 t = [1e-6; 1e-7];
8 ang = (1/omSq)*[i(3,3)-i(2,2) i(1,2); ...
9               i(3,3)-i(1,1) i(1,3)]; ...
10      \[ t + omSq*[i(2,3); -i(1,3)]]

1 omega = 0.0011
2 ang = 1.0e-07 *[0.0047; 0.1046]
```

Example 13.8: Gravity-gradient momentum unloading.